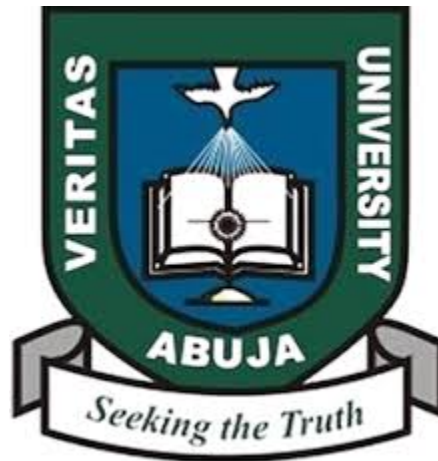


Department of Software Engineering  
Veritas University, Bwari Abuja.



‘LAB MANUAL’  
Course Code: VUA-SEN182  
Course Title: “Web Programming (HTML, CSS, Git)”

**Prepared by: Mr. Victor Ayamba**

## Course Contents

| Contents                                                                                                                                                                                                                                                                              | CLOs        | Lecture | Practical |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------|-----------|
| HTML: Basics, Elements, Attributes, Headings, Paragraphs, Formatting, Links, Head, Images, Tables, Lists, Blocks, Layout, Forms, Multimedia Tags, Semantic HTML, HTML5 Structure.                                                                                                     | CLO – 1 & 2 | 4       | 8         |
| CSS: Introduction, Syntax, Selectors (ID & Class), Styling Backgrounds, Text, Fonts, Links, Box Model (Border, Outline, Margin, Padding), Grouping/Nesting, Dimension, Display, Positioning, Floating, Align, Navigation Bar, Image Gallery, Flexbox, Grid Layout, Responsive Design. | CLO – 1 & 2 | 3       | 6         |
| JavaScript: Introduction, Output, Statements, Comments, Variables, Data Types, Objects, Functions, Operators, Comparisons, Conditions, Loops, Error Handling, DOM Introduction and Manipulation, Number, String, Date, Array, Window, Screen, Navigator, Popup Alerts.                | CLO – 1 & 2 | 4       | 8         |
| Bootstrap: Introduction to Bootstrap Framework, Grid System, Containers and Layout, Responsive Design Principles, Typography and Utilities, Navigation Bars, Buttons, Cards, Forms and Input Groups, Tables, Alerts, Modals, Responsive Components and Layout Customization.          | CLO – 1 & 2 | 4       | 8         |
| Total                                                                                                                                                                                                                                                                                 |             | 15      | 30        |

## INTRODUCTION TO THE COURSE:

The Internet and the World Wide Web have become fundamental components of modern society, influencing communication, education, business, and everyday life. As digital platforms continue to expand, the ability to design and develop functional and visually appealing web applications has become an essential skill for software engineers and technology professionals. Understanding how web technologies work together enables students to create interactive systems that can serve real-world needs.

This laboratory course provides a practical introduction to modern Web Programming using core client-side technologies such as **HTML, CSS, JavaScript, and the Bootstrap framework as well as version control tools like Git and Github**. Students will learn how to structure web pages using HTML, style and layout interfaces with CSS, enhance interactivity through JavaScript, and build responsive, mobile-friendly user interfaces using Bootstrap. Through hands-on laboratory exercises, students will progressively develop the skills required to design, implement, and maintain modern web applications.

The lab sessions are structured to guide students step-by-step through the development of a meaningful web project while introducing essential web development concepts and tools. Each experiment builds upon the previous one, enabling learners to gradually transform basic web pages into fully structured and responsive websites. At the end of each lab session, tasks are provided to reinforce understanding and encourage independent exploration. Students are advised to follow this lab manual sequentially and carefully adhere to the provided guidelines to gain maximum benefit from the course.

## LAB 1: Introduction to Web Programming

### SECTION A: Tutorial - Demonstrated by Instructor

#### Experiment Name: Installation of Visual Studio Code.

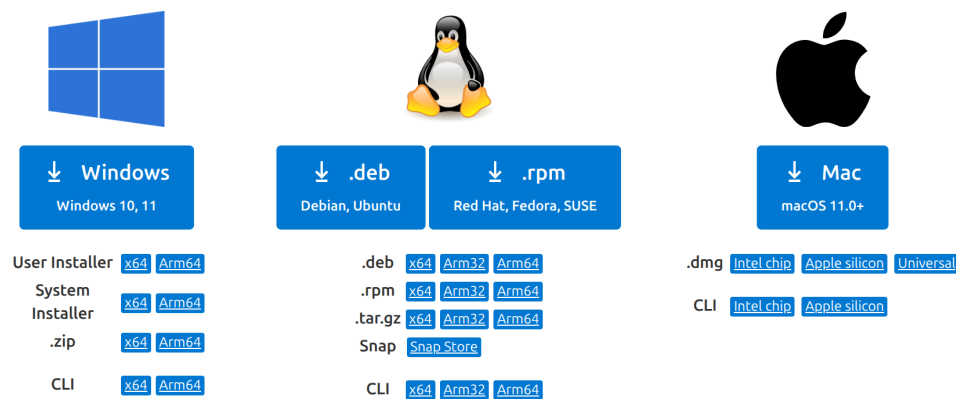
**Theory:** Visual Studio Code, commonly referred to as VS Code, is a source-code editor made by Microsoft with the Electron Framework for Windows, Linux, and macOS. It features support for debugging, syntax highlighting, intelligent code completion, snippets, code refactoring, and embedded Git.

**Procedure:** Follow the steps below to install VS Code on Windows:

**Step 1:** Visit <https://code.visualstudio.com> and click on the "Windows Download" option. By clicking the download option, an exe file (approximately 100 MB) will be downloaded, and the process should not take much time.

## Download Visual Studio Code

Free and built on open source. Integrated Git, debugging and extensions.



By downloading and using Visual Studio Code, you agree to the [license terms](#) and [privacy statement](#).

Figure 1: Download Interface of VS code

**Step 2:** After the file is downloaded, open it, and the following screen will appear. Check the "I accept the agreement" option and proceed by clicking the "Next" button.

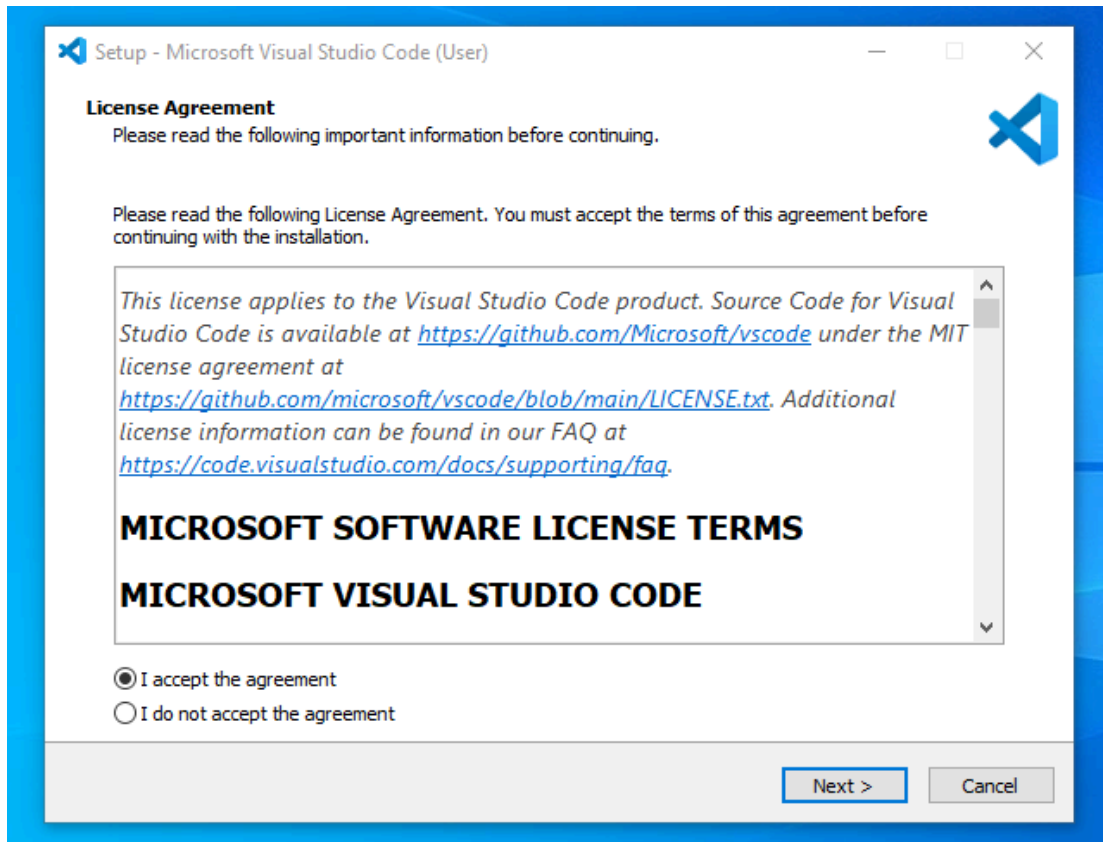


Figure 2: Installation Process of VSCode

**Step 3:** The next screen is for selecting the file location. You can avoid changing the default location by simply clicking the "Next" button.

**Step 4:** Choose the location data for running the Visual Studio Code. It will then ask you to browse the location. Then click on the Next button. This is optional

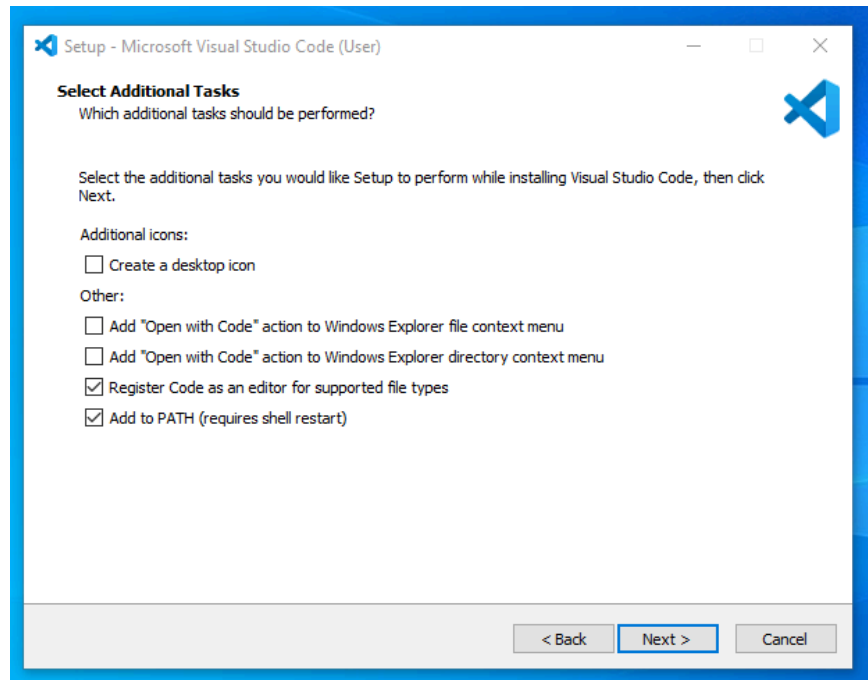


Figure 3: Installation Process

**Step 5:** Then it will ask to begin the installation setup. Click on the Install button.

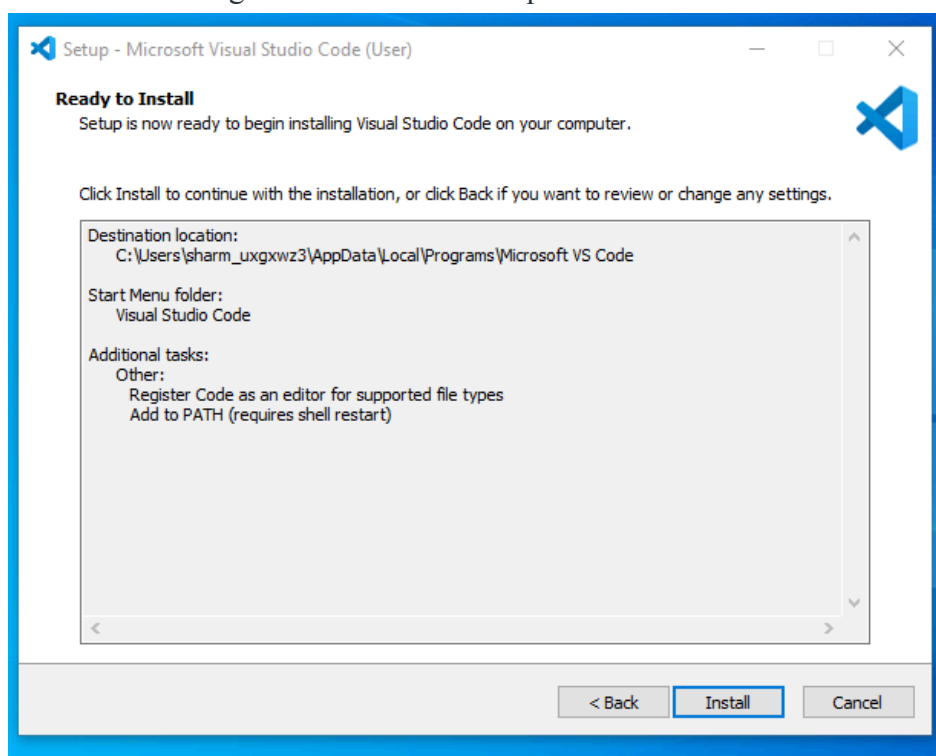


Figure 4: VS Code Installation process Continues

**Step 6:** After clicking on Install, it will take about 1 minute to install the Visual Studio Code on your device.

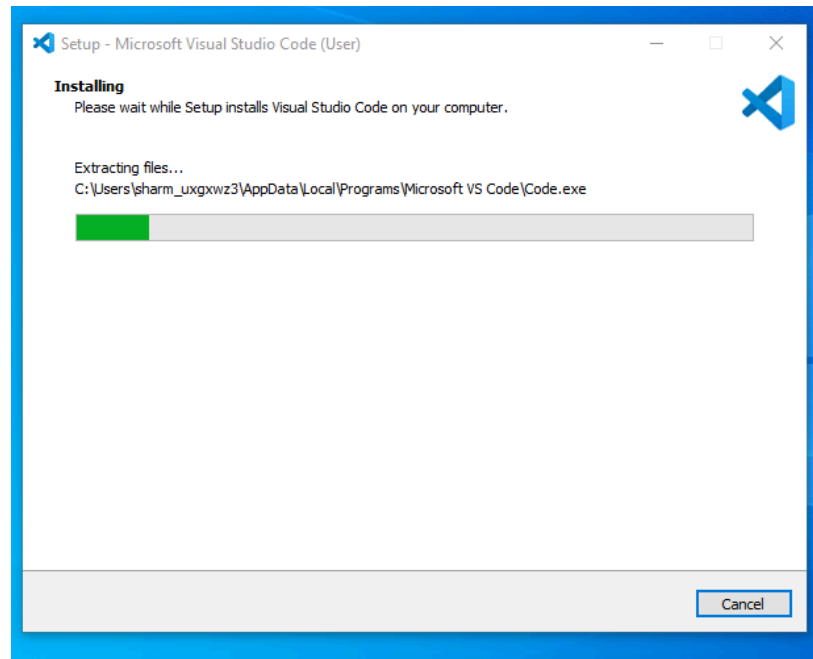


Figure 5: VS Code Installation Process Continues..

**Step 7:** After the Installation setup for Visual Studio Code is finished, it will show a window like this below. Tick the "Launch Visual Studio Code" checkbox and then click Next.

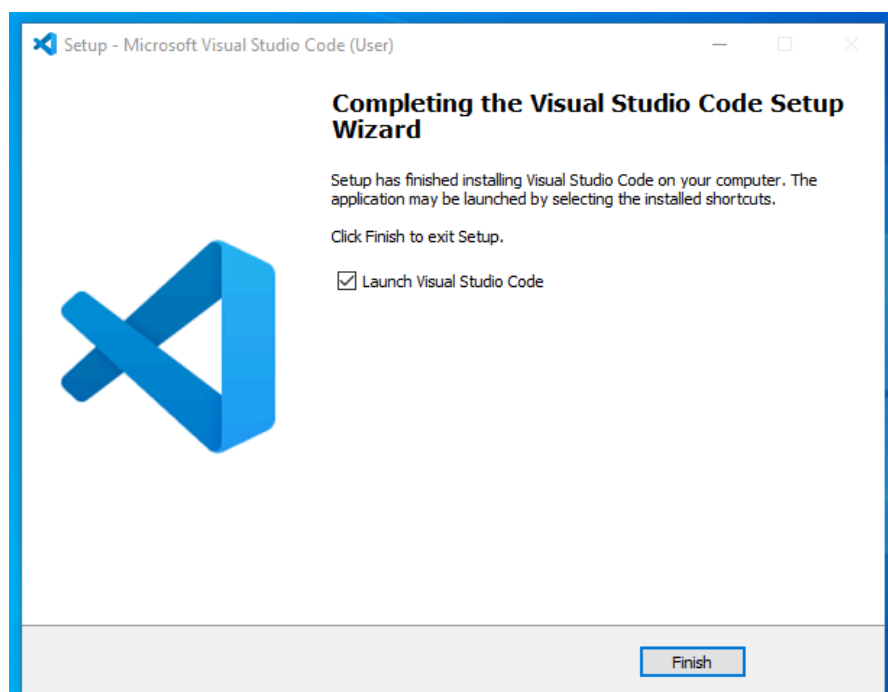


Figure 6: VS Code Installation process Continues.

**Step 8:** After the previous step, the Visual Studio Code window opens successfully. Now you can create a new file in the Visual Studio Code window and choose a language of yours to begin your programming journey!

**Conclusion:** In conclusion, the installation process for Visual Studio Code on Windows is straightforward and user-friendly. By following a series of simple steps, you can quickly download, install, and set up the Software.



## Lab 2: Introduction to HTML and Basic Web Page Structure

### SECTION A: Tutorial - Demonstrated by Instructor

**Experiment Name:** HTML Fundamentals; Document Structure, Text Elements, Links, Images and Basic Styling

#### Theory

HTML (**HyperText Markup Language**) is the standard language used to create and structure web pages on the Internet. It defines the structure and layout of content such as headings, paragraphs, images, links, tables, lists, and multimedia. HTML works together with CSS and JavaScript to create modern interactive websites.

In this laboratory session, students will learn the **fundamental building blocks of HTML**, including how to create a basic HTML document, structure content using headings and paragraphs, add images and hyperlinks, and apply simple styling.

Every HTML document follows a standard structure and is interpreted by web browsers such as **Google Chrome, Microsoft Edge, Firefox, and Safari**.

#### Creating an HTML File

To create and run an HTML document:

1. Open a code editor such as **Visual Studio Code**, Notepad++, or Sublime Text.
2. Create a new file.
3. Save the file with the extension **.html** (example: [index.html](#)).
4. Open the file using a web browser.

#### HTML Document Structure

Every HTML page begins with a **document type declaration** followed by the HTML structure.

#### Example

```
<!DOCTYPE html>
<html>
<body>

<h1>My First Heading</h1>
<p>My first paragraph.</p>

</body>
</html>
```

#### Explanation of the Structure

| Element         | Description                                  |
|-----------------|----------------------------------------------|
| <!DOCTYPE html> | Declares that the document is HTML5          |
| <html>          | Root element of the HTML document            |
| <body>          | Contains the visible content of the web page |
| <h1>            | Defines a main heading                       |
| <p>             | Defines a paragraph                          |

### The Declaration

The <!DOCTYPE html> declaration tells the browser which version of HTML is being used.

Important notes:

- It must appear **at the top of the HTML document**.
- It appears **only once** in the document.
- It ensures browsers render the page correctly.

Example:

<!DOCTYPE html>

### HTML Headings

Headings are used to define titles and subtitles on a web page.

HTML provides **six heading levels**:

| Tag  | Description             |
|------|-------------------------|
| <h1> | Most important heading  |
| <h2> | Second level heading    |
| <h3> | Third level heading     |
| <h4> | Fourth level heading    |
| <h5> | Fifth level heading     |
| <h6> | Least important heading |

Example:

<h1>Heading 1</h1>  
<h2>Heading 2</h2>  
<h3>Heading 3</h3>

### Important Notes

- <h1> should normally appear **once per page**.
- Headings help **search engines understand the structure of a page**.

Example Structure:

<h1>Travel Guide</h1>

<h2>Europe</h2>

<h3>France</h3>

<h3>Italy</h3>

<h2>Asia</h2>

<h3>India</h3>

<h3>Thailand</h3>

<h2>North America</h2>

<h3>United States</h3>

### HTML Paragraphs

Paragraphs are defined using the <p> tag.

Example:

<p>This is a paragraph.</p>

<p>This is another paragraph.</p>

Important characteristics:

- A paragraph always starts on a **new line**.
- Browsers automatically add **spacing before and after paragraphs**.

### HTML Horizontal Lines

The <hr> tag defines a thematic break in content.

Example:

<h1>This is heading 1</h1>

<p>This is some text.</p>

<hr>

<h2>This is heading 2</h2>

<p>This is another text.</p>

## HTML Line Breaks

The <br> tag inserts a line break.

Example:

<p>This is a paragraph<br>with line breaks.</p>

## The <pre> Element

The <pre> tag displays text exactly as written, preserving spaces and line breaks.

Example:

```
<pre>
My Bonnie lies over the ocean
My Bonnie lies over the sea
My Bonnie lies over the ocean
Oh bring back my Bonnie to me
</pre>
```

## HTML Elements

An HTML element is defined by:

Start tag + Content + End tag

Example:

<h1>My First Heading</h1>

| Part             | Description |
|------------------|-------------|
| <h1>             | Start tag   |
| My First Heading | Content     |
| </h1>            | End tag     |

Some elements do not require closing tags. These are called **empty elements**.

Example:

<br>

<hr>

## HTML Attributes

Attributes provide **additional information about elements**.

Attributes are always placed **inside the opening tag**.

Example:

<a href="https://www.example.com">Visit Website</a>

| Attribute | Description                   |
|-----------|-------------------------------|
| href      | Specifies the destination URL |

## HTML Images

Images are inserted using the <img> tag.

Example:



| Attribute | Purpose                            |
|-----------|------------------------------------|
| src       | Image file path                    |
| alt       | Alternative text for accessibility |
| width     | Image width                        |
| height    | Image height                       |

There are two types of image paths:

### Absolute URL

https://www.example.com/image.jpg

### Relative URL

images/photo.jpg

## HTML Links (Hyperlinks)

Links allow navigation between web pages.

Example:

```
<a href="https://www.w3schools.com">Visit W3Schools</a>
```

### Opening a link in a new tab

```
<a href="https://www.w3schools.com" target="_blank">
Visit W3Schools
</a>
```

### Link to Email

```
<a href="mailto:someone@example.com">Send Email</a>
```

## HTML Text Formatting

HTML provides tags for formatting text.

| Tag      | Description      |
|----------|------------------|
| <b>      | Bold text        |
| <strong> | Important text   |
| <i>      | Italic text      |
| <em>     | Emphasized text  |
| <small>  | Smaller text     |
| <mark>   | Highlighted text |
| <del>    | Deleted text     |
| <ins>    | Inserted text    |
| <sub>    | Subscript        |
| <sup>    | Superscript      |

Example:

```
<p>This text is <strong>important</strong>.</p>
```

<p>This text is <em>emphasized</em>.</p>  
<p>This text is <mark>highlighted</mark>.</p>

## HTML Quotations and Citations

HTML includes elements for quotations and references.

| Tag          | Description         |
|--------------|---------------------|
| <blockquote> | Long quotation      |
| <q>          | Short quotation     |
| <abbr>       | Abbreviation        |
| <address>    | Contact information |
| <cite>       | Title of work       |
| <bdo>        | Text direction      |

Example:

<blockquote>

This is a quotation from another source.

</blockquote>

<p>WWF's goal is <q> Build a future where people live in harmony with nature.</q></p>

## HTML Comments

Comments are used to document code and are not displayed in browsers.

**Syntax:**

<!-- This is a comment -->

Example:

<!-- Website Header -->

<h1>My Website</h1>

## HTML Colors

Colors can be applied using the **style attribute**.

Example:

<h1 style="color:blue;">Hello World</h1>

<p style="color:red;">This is a paragraph</p>

## Color Formats

| Format     | Example         |
|------------|-----------------|
| Color Name | red             |
| RGB        | rgb(255,0,0)    |
| HEX        | #ff0000         |
| HSL        | hsl(0,100%,50%) |

## Introduction to CSS Styling

CSS (Cascading Style Sheets) controls the appearance of web pages.

CSS can be applied in three ways.

### 1 Inline CSS

```
<h1 style="color:blue;">Heading</h1>
```

### 2 Internal CSS

```
<style>
h1 {color: blue;}
</style>
```

### 3 External CSS

```
<link rel="stylesheet" href="style.css">
```

Example external CSS file:

```
body {
background-color: powderblue;
}

h1 {
color: blue;
}
```



## HTML Links Styling with CSS

Links can be styled using CSS states.

Example:

```
a:link {  
  color: green;  
}
```

```
a:hover {  
  color: red;  
}
```

```
a:visited {  
  color: purple;  
}
```

## Lab 2: Introduction to HTML and Basic Web Page Structure

### SECTION B: Lab Tasks

Your task is to build a simple **Orientation Information Webpage for Veritas University**. The goal is to create a simple webpage that provides useful information to newly admitted students. Each task below applies the HTML concepts learned in this lab session.

#### Task 1: Create the Basic HTML Document Structure

Create a new HTML file called **orientation.html**.

Your file must include the basic HTML document structure:

- `<!DOCTYPE html>`
- `<html>`
- `<head>`
- `<title>`
- `<body>`

#### What to do

Write the HTML skeleton of your webpage.

#### Concept Applied

- HTML document structure
- DOCTYPE declaration
- Root element `<html>`

#### Task 2: Create the Main Page Heading

Add a heading that welcomes new students.

#### What to do

Inside the `<body>` section, add a **main heading** for the page.

#### Suggestion

`<h1>Welcome to Veritas University Freshman Orientation Portal</h1>`

You may also add a secondary heading describing the purpose of the page.

Example:

<h2>Important Information for Newly Admitted Students</h2>

### Task 3: Add Introduction Paragraphs

Provide a short explanation about the purpose of the orientation page.

#### What to do

Add **two paragraphs** describing:

- The purpose of the webpage
- How it helps new students

#### Suggestion

<p>This webpage provides important information to guide newly admitted students during their orientation period at Veritas University.</p>

<p>Here you will find details about registration procedures, orientation schedules, important contacts and useful university resources.</p>

### Task 4: Add Orientation Information Section

Provide key orientation information using headings and paragraphs.

#### What to do

Add a section explaining:

- Orientation date
- Required documents
- Registration instructions

#### Suggestion

<h2>Orientation Information</h2>

<p><strong>Orientation Date:</strong> 15th September 2026</p>

<p><strong>Venue:</strong> Veritas University Main Auditorium</p>

<p>Students are required to come along with admission letters, identification documents and proof of fee payment.</p>

## Task 5: Insert an Image

Add an image that represents the university.

### What to do

Insert an image such as:

- Veritas University campus
- Orientation banner

### Suggestion

```

```

## Task 6: Create Useful Links for Students

Add links to important university resources.

### What to do

Create hyperlinks to:

- University website
- Student portal
- Email contact

### Suggestion

```
<h2>Useful Resources</h2>
```

```
<p>
```

```
<a href="https://www.veritas.edu.ng">Visit Veritas University Website</a>
```

```
</p>
```

```
<p>
```

```
<a href="mailto:support@veritas.edu.ng">Contact Student Support</a>
```

```
</p>
```

## Task 7: Emphasize Important Notices

Highlight important instructions using formatting tags.

### What to do

Add at least **three formatted text elements**.

### Suggestion

*<p><mark>Note:</mark> All students must complete registration before attending lectures.</p>*

*<p><em>Late registration may attract penalties.</em></p>*

*<p><small>Please arrive at least 30 minutes before the orientation begins.</small></p>*

### Task 8: Add a Welcome Message Using a Quotation

Include a motivational message for new students.

### What to do

Use a quotation tag to display a message.

### Suggestion

*<blockquote>*

*Education is the most powerful tool you can use to change the world. Welcome to Veritas University.*

*</blockquote>*

### Task 9: Add Contact Information

Provide contact information for student support.

### What to do

Use the *<address>* tag.

### Suggestion

*<h2>Contact Information</h2>*

*<address>*

*Student Affairs Unit<br>*

*Veritas University, Abuja<br>*

*Email: support@veritas.edu.ng<br>*

*Phone: +234 800 000 0000*

*</address>*

### **Task 10: Add HTML Comments**

Add comments to describe sections of your code.

#### **What to do**

Insert at least **two comments**.

#### **Suggestion**

*<!-- Orientation Information Section -->*

*<!-- Student Support Contact Section -->*

### **Expected Output**

Your webpage should display:

- A welcome heading
- Orientation information
- A university image
- Useful links
- Highlighted notices
- A welcome quotation
- Contact information

A sample of the web page is displayed below

# Welcome to Veritas University Freshman Orientation Portal

## Important Information for Newly Admitted Students

This webpage provides important information to guide newly admitted students during their orientation period at Veritas University.

Here you will find details about registration procedures, orientation schedules, useful resources, and contact information for support.

### Orientation Information

**Orientation Date:** 15th September 2026

**Venue:** Veritas University Main Auditorium

Students are expected to come along with their admission letter, proof of school fees payment, passport photographs, and valid means of identification.

### Registration Steps

This section outlines the key steps every new student must complete:

1. Complete online acceptance and school fee payment
2. Print all payment receipts and admission documents
3. Attend departmental clearance
4. Proceed to faculty and central registration units

### Campus Preview

Below is a sample image representing the university environment:

 Veritas University Campus

### Useful Resources

[Visit Veritas University Website](#)

## References & Further studies

1. [https://www.w3schools.com/html/html\\_css.asp](https://www.w3schools.com/html/html_css.asp)
2. <https://developer.mozilla.org/en-US/docs/Web/HTML>
3. <https://www.geeksforgeeks.org/html/html-tutorial/>